

Clinical and Oral Microbiology: Introduction to Virology

Integrated Biomedical and Clinical Sciences OMFS 513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Objectives

1. Describe the virus composition and classification.
2. Describe the viral life cycle and transmission modes.
3. Describe viral hepatitis.
4. Explain human immunodeficiency virus.

Outline

- Foundations:
 - Viral structure
 - Life cycle
 - Transmission
- Medical virology
 - Oncogenic viruses
 - Adenovirus
 - Viral hepatitis
 - Respiratory Syncytial virus
 - Human immunodeficiency virus

What is the virus?

(A virus is) ?

“simply a piece of bad news wrapped in a protein coat.” — Peter Medawar, 1977.

Virus: Latin for poisonous liquid



Sir Peter Medawar, British biologist, Nobel Prize winner, physiology/medicine, 1960

[Bad News Wrapped in Protein: Inside the Coronavirus Genome - The New York Times \(nytimes.com\)](#)

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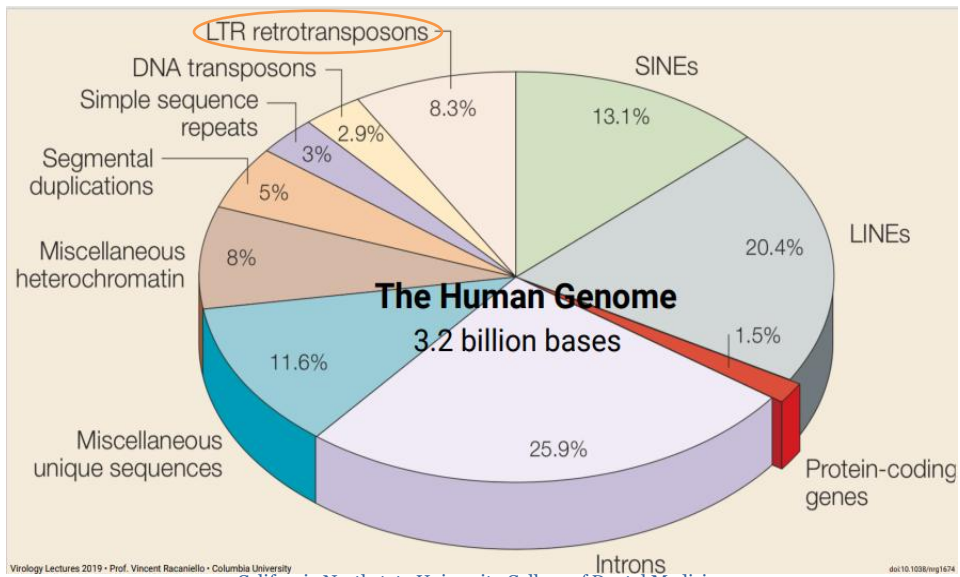
Types of Viruses

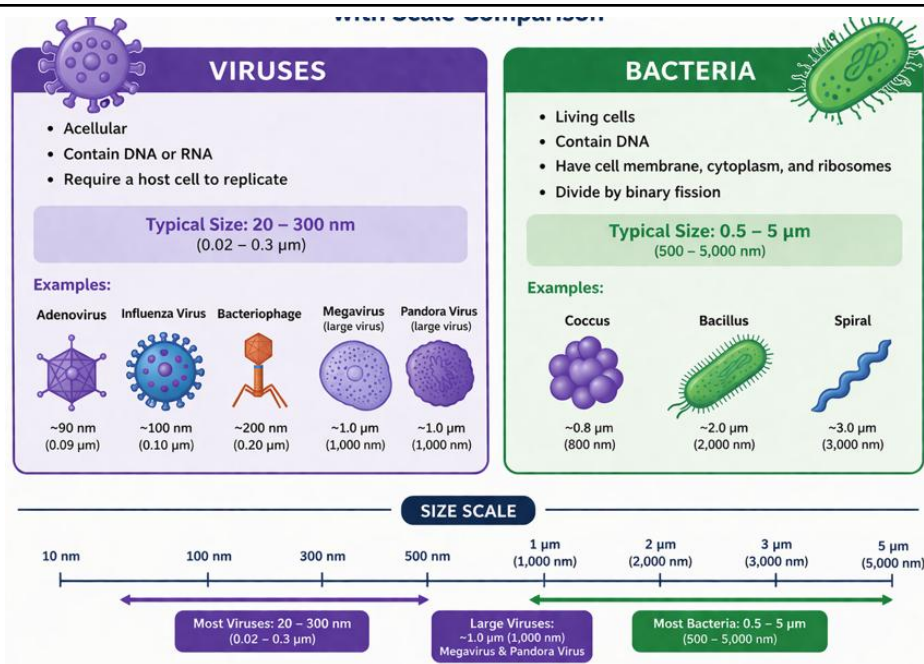
- Animal and human viruses
- Plant viruses
- Insect viruses
- Bacterial viruses: bacteriophage
- Fungal viruses

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8 % of our genome comes from ancient Viral DNA!!





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Basic Characters

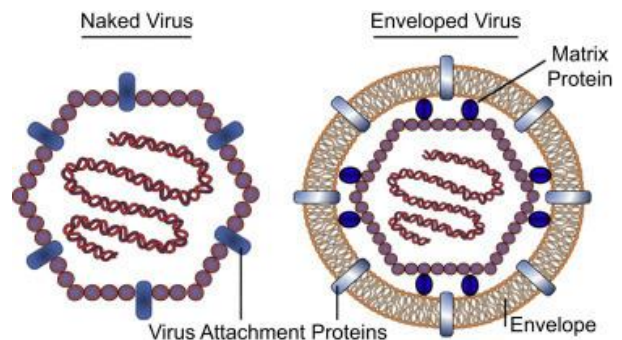
- 1- Infectious parasite
- 2- Lack their own metabolism
- 3- Replicate inside a cell (prokaryote or eukaryote)
- 4- Nucleic acid + surrounding shell.
- 5- Capsid/envelop are for protection, attachment & determine tropism (host range).

Architecture

A virus is an infectious intracellular parasite.

Composed of:

- Nucleocapsid (genome + capsid). Genetic material (either DNA or RNA), packaged in a protein-coat named capsid.
- With or without an envelope.



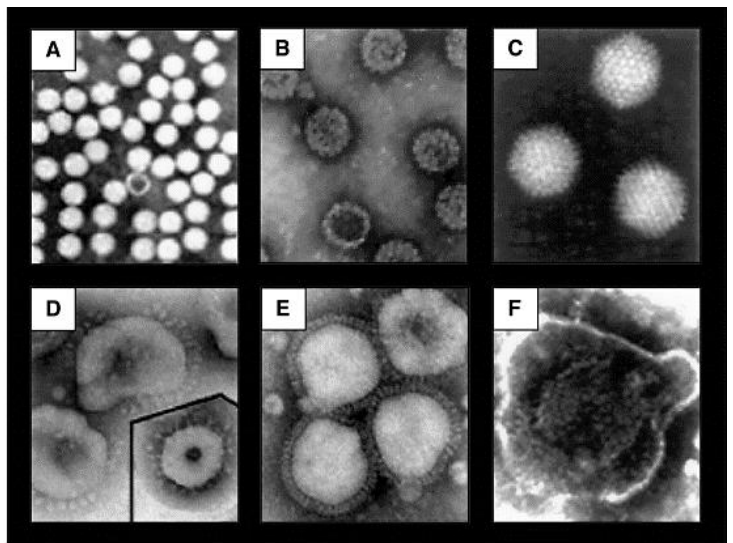
Envelope

- Viruses are either naked (e.g Hep A/E) or enveloped (Hep B/C; HIV and herpes).
- Viral envelope surrounds the nucleocapsid.
- Composed of viral glycoproteins and host-cell-derived lipoproteins.

Functions:

- Help in viral binding to the host cell to initiate infection.
- Can stimulate antibody production.
- Can serve as antigen in serologic testing.
- Sensitive to chemical and physical treatment

Naked (a-c) & enveloped (D-f) viruses.

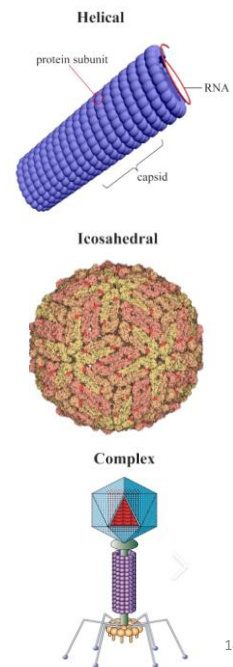


Viral capsid

- Made up of viral proteins arranged in structural units named capsomers.
- According to capsid shape, viral architecture is classified to helical, icosahedral (a 20-sided polygon), or complex.

Capsid functions:

- Protect the viral genome.
- Constitute receptor sites necessary for initiating infection in naked virus.
- Antigenic.



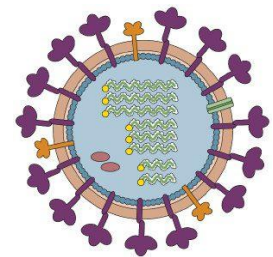
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Viral genome

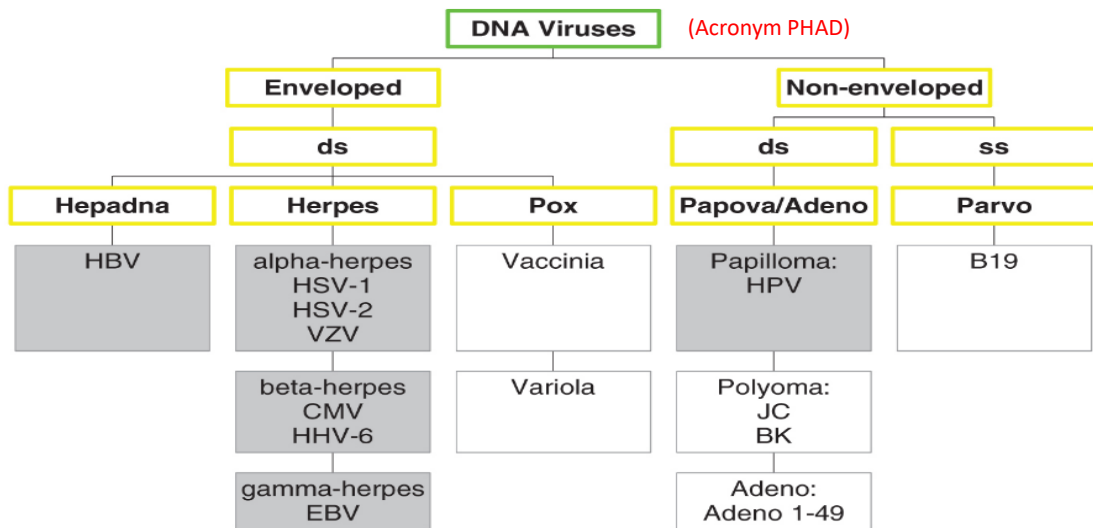
Either linear or circular. DNA or RNA.

- DNA virus:
 - Single stranded
 - Double stranded
- RNA virus:
 - Non-segmented
 - Single stranded RNA with positive (+) or negative (-) polarity.
 - Double stranded
 - Segmented
 - Can also be single or double stranded. Example: influenza virus
 - Allows re-assortment of genome.



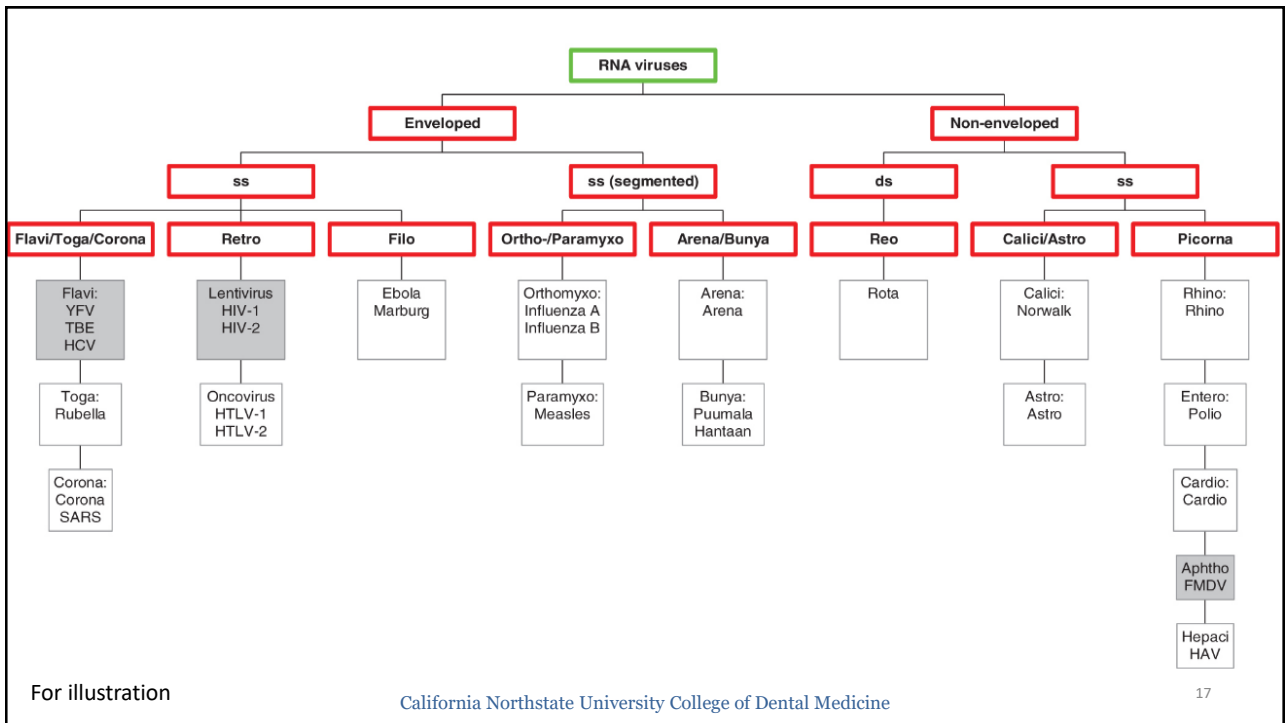
Segmented RNA virus

Virl Taxonomy



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Life cycle and replication

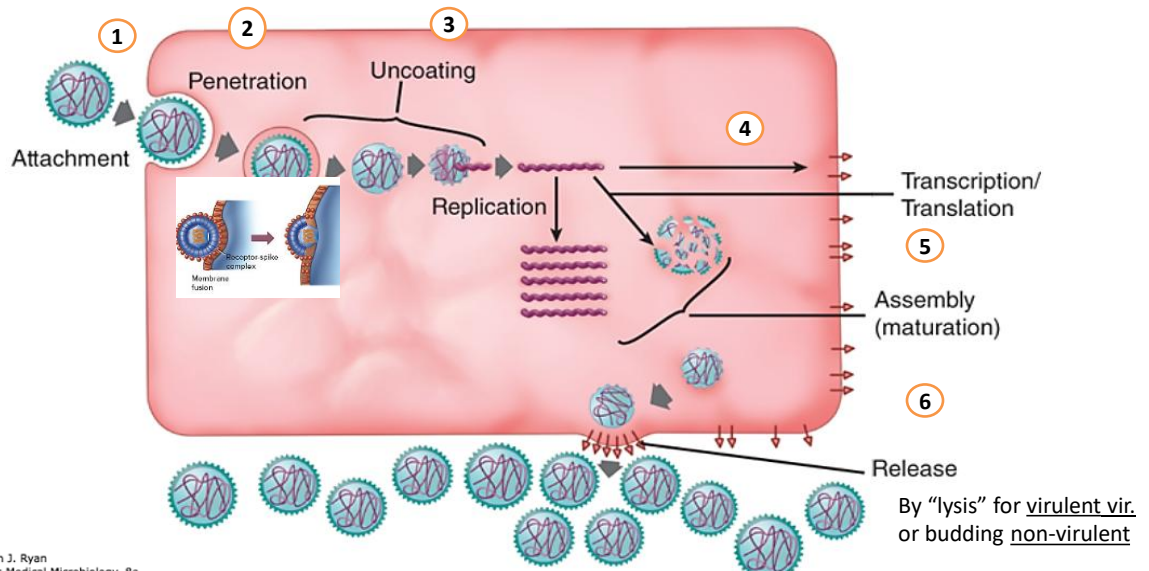
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Phases of Life Cycle

- Intracellular form
- Extracellular form “virion”

Replication Cycle



Source: Kenneth J. Ryan
 Sherris & Ryan's Medical Microbiology, 8e
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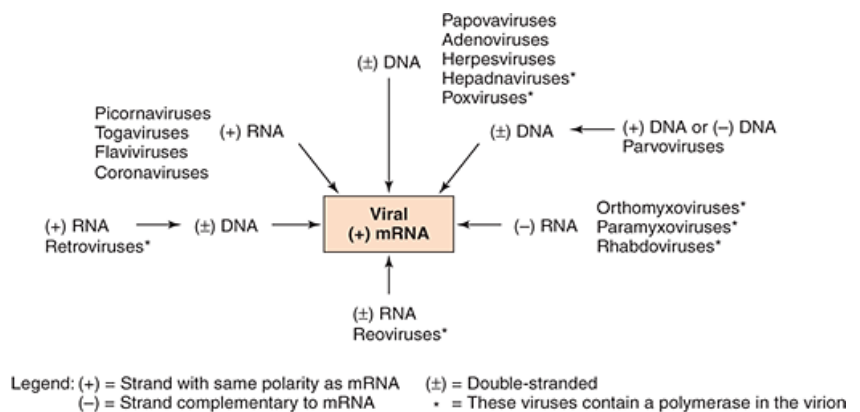
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- 1) Adsorption or attachment** to the host cell. Attachment of the viral attachment proteins (VAPs) or spikes to the host cell receptors. The presence of compatible host cell receptors determines the susceptibility of the cells. E.g, the interaction between the SARS-CoV-2 spike protein and the human angiotensin-2.
- 2) Penetration or entry.** This involves the fusion of the viral envelope with the cell membrane or the viral uptake into the cells via endocytosis or fusion with membrane.
- 3) Uncoating** to release the viral genome. The viral capsid and genetic material are released into the cytoplasm of the host cell.
- 4) Synthetic phase** (transcription, translation and replication). Depends on the viral genome
- 5) Assembly** The progeny particles are assembled by packaging the viral nucleic acid within the capsid proteins.
- 6) Viral release from host cell.** One infected cell can make thousands of virion particles. Those virions can immediately infect near by cells.

Synthetic Phase (genome replication + protein production)

All viruses must make/have mRNA for synthetic phase

Baltimore
classification is used
to classify viruses
based on manner of
messenger RNA
(mRNA) synthesis



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- DNA viruses: Replicate in nucleus (except for poxviruses)
 - Class I dsDNA virus → mRNA
 - Class II ssDNA → dsDNA → mRNA
- RNA viruses: mainly replicate in cytoplasm (except influenza viruses and retroviruses).
 - Class III dsRNA → mRNA
 - Class IV/V ssRNA → mRNA
- Additional classes
 - Retrovirus (Class VI) RNA to DNA then RNA
 - Hep B (Class VII; DNA virus with RNA intermediate)

Synthetic Phase for DNA virus

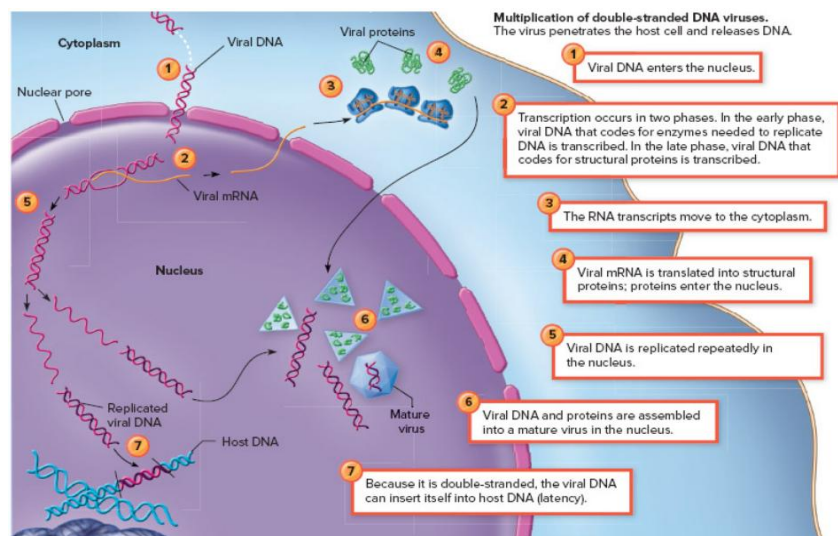
Transcription regulated by host-DNA-dependent RNA polymerases (to yield mRNA).

Translation occurs in cytoplasm and newly synthesized proteins go to the nucleus (except poxviruses)

Replication occurs in the nucleus after protein synthesis using host or viral polymerase.

Assembly in the nucleus (except poxviruses).

In some cases, viral DNA can integrate in host DNA (provirus).



Synthetic phase for RNA virus

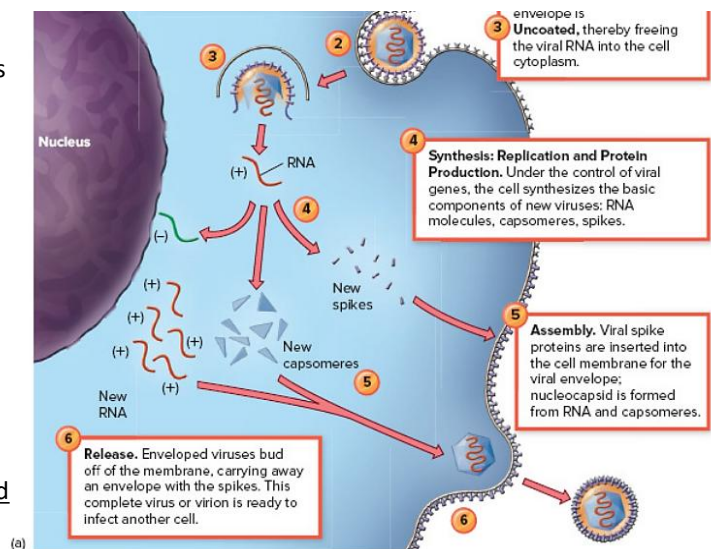
a) **Positive sense ss RNA viruses:** can immediately be translated to produce viral proteins as well as RNA polymerase and start replication.

a) **Negative sense ss RNA viruses.** uses their viral RNA-dependent RNA polymerase for transcription (aka, RNA transcriptase) to generate +ss mRNA.

Translation occurs in the cytoplasm.

Genome replication occurs in the cytoplasm (except for retroviruses) and is performed by a viral-specific replicase enzyme (except for retroviruses).

RNA viruses have a higher mutation rate compared to DNA viruses due to the lack of polymerase proofreading ability.



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Assembly and Release

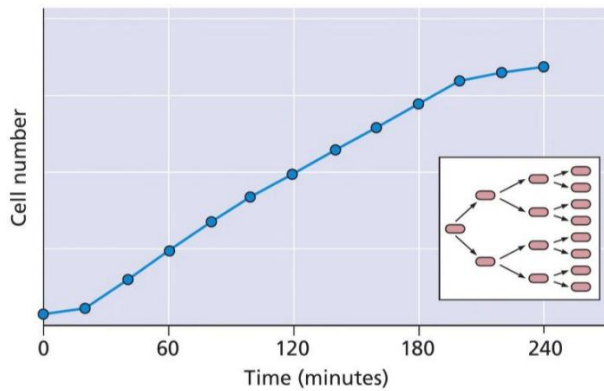
Assembly: mature virus particles are constructed from the constructed parts. Most enveloped viruses are assembled in the cytoplasm via matrix proteins. Naked DNA viruses are assembled in the nucleus while naked RNA viruses are in the cytoplasm.

Release:

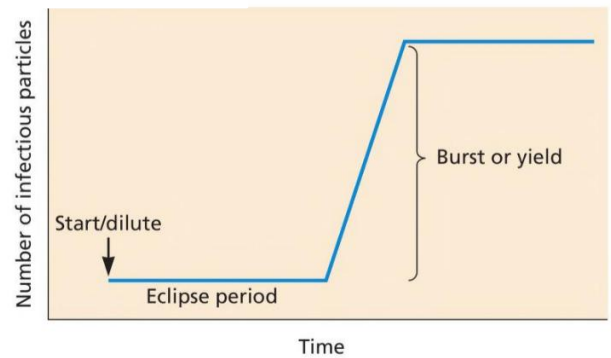
- Nonenveloped & complex viruses are released through cell lysis or rupturing.
- Enveloped viruses are released by budding or exocytosis. from the cell membrane. Virus acquires a part of host cell membrane.

Bacterial vs Viral replication

A Bacteria



B Virus (bacteriophage)



Transmission and Pathogenesis

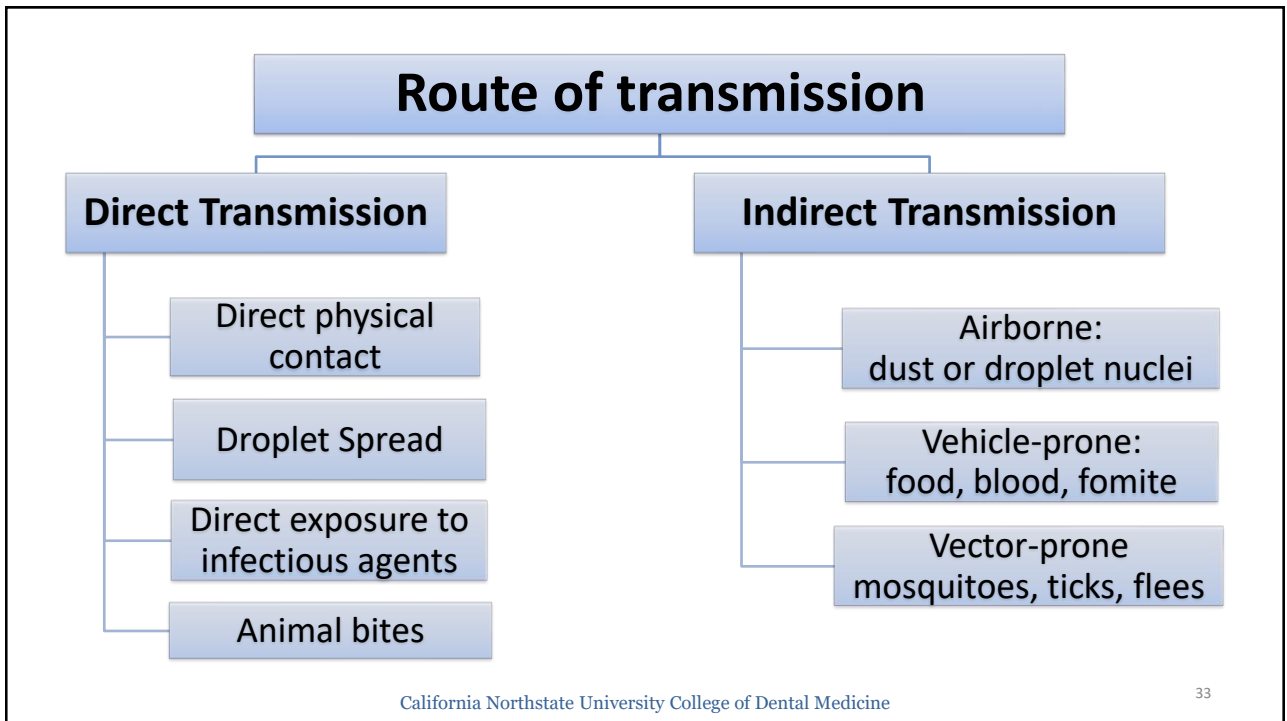
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Pathogenesis in the Infected Patient

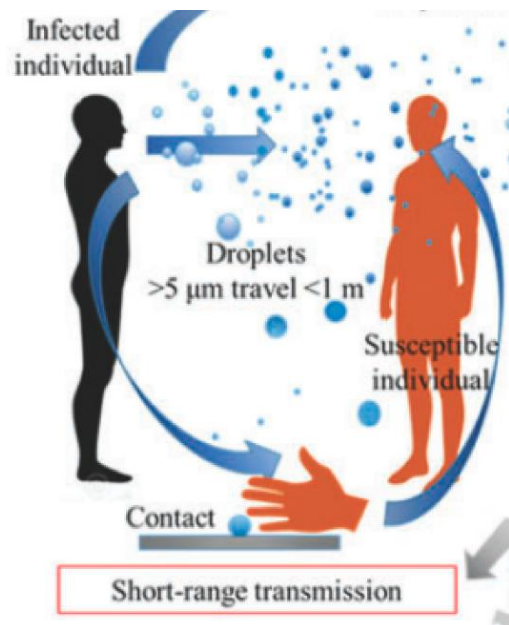
In the host, viral infection involve:

- (1) Exposure and transmission of virus.
- (2) Interactions of virus with cells and viral replication “viral tropism”.
- (3) Localized or spread of the virus to other cells/organs.
- (4) Innate and adaptive immune responses as well as viral ability to evade the immune system.
- (5) Viral persistence in some cases.



A) Direct: pathogen is transferred from a reservoir to a susceptible host via:

- I. **Direct physical contact:** like, infectious mononucleosis ("kissing disease") spread by direct contact. Hookworm is spread by direct contact with contaminated soil.
- II. **Droplet Spread:** medium to large-sized aerosol particles produced by sneezing or coughing or talking.



B) Indirect: transfer of the pathogen from the reservoir to the host via:

- I. **Airborne:** suspended viral particles in droplet nuclei (SARS-CoV-2)
- II. **Vehicle-borne** like food, water, blood and fomites. Vehicles can either passively carry the pathogen (e.g: contaminated food can harbor hepatitis A virus) or a vehicle can provide an environment that allows the growth of pathogen or toxin production (e.g: *Clostridium botulinum*).
- III. **Vector-borne:** vectors may carry pathogens, or they may also support the pathogen growth (e.g: mosquitoes help with spreading and completing the malaria life cycle).

Modes of Transmission

Horizontal transmission: between members of same species.

Vertical transmission: between mother and child.

Iatrogenic: between health care worker to patient.

Nosocomial: infections in a hospital or health care facility.

Germ-line transmission: transmission through the genome.

Viral Tropism

- **Susceptible cell** has a functional receptor for that promote viral entry.
- **Permissive cell** has the capacity to replicate virus. It may or may not be susceptible to the virus.
- **Susceptible and permissive cell** are the only cells that can uptake and replicate a virus particle.
- **Resistant cell** has no receptor, and it may or may not be capable of supporting the viral replication leading to abortive infection.

Types of infection

1- **Inapparent infections or subclinical disease.** Number of infected cells is insufficient to display clinical symptoms. Can be associated with sufficient immune response to prevent the spread of infection.

2. **Acute infections.** Clinical manifestations are observed for a short period of time (days to weeks). They are also accompanied with a short incubation period. In some cases, persistent or latent infections may follow the acute infection phase.

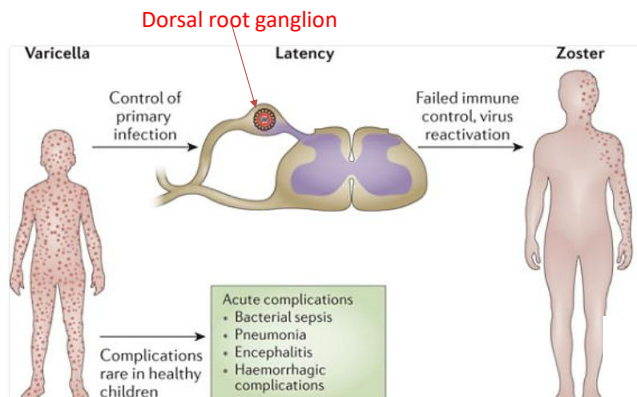


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3. Latent infections or recurrent disease

In this case, virus persists in the body in a noninfectious stage. Can reactivate to an infectious virus and produce clinical symptoms. Example: herpes family. Like, varicella zoster virus (HHV-3) remains latent in nerve ganglia and reactivate to produce shingles.



4. Chronic infection are associated with continued presence of the infectious virus in the body for an extended period, examples include HBV, HCV.

5. Slow-virus infections.

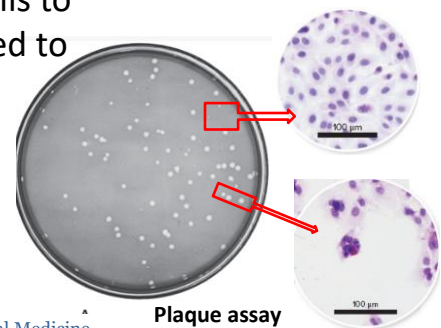
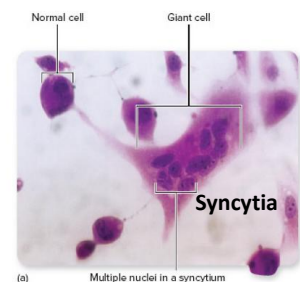
Those have a prolonged incubation period lasting months or years. They are most often associated with chronic, progressive, fatal viral diseases such as HIV and some lentivirus infections.



Pathogenesis of Infected cells

Virus infection can lead to:

- No apparent morphologic or functional change.
- Cytopathic effects (CPE) define the change in the appearance of the infected cell. It can include cell rounding, fusion of cell membrane of multiple cells to form multinucleated cell. CPE changes can be used to identify the type of infection.
- Malignant transformation (oncogenic virus).
- Cell death (plaques).



[VitalSource Bookshelf: Microbiology: A Systems Approach](#)

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Diagnosis- Management & Prevention

- Diagnosis: Clinical diagnosis, detection of antigen or antibody (serology), detection of viral genes
- Management: anti-viral agents. Virus specific (unique viral enzyme/replication component); liable to resistance.
- Prevention: Infection control & vaccines.
- Post-exposure protocols

Viruses Causing Human Diseases

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Oncoviruses

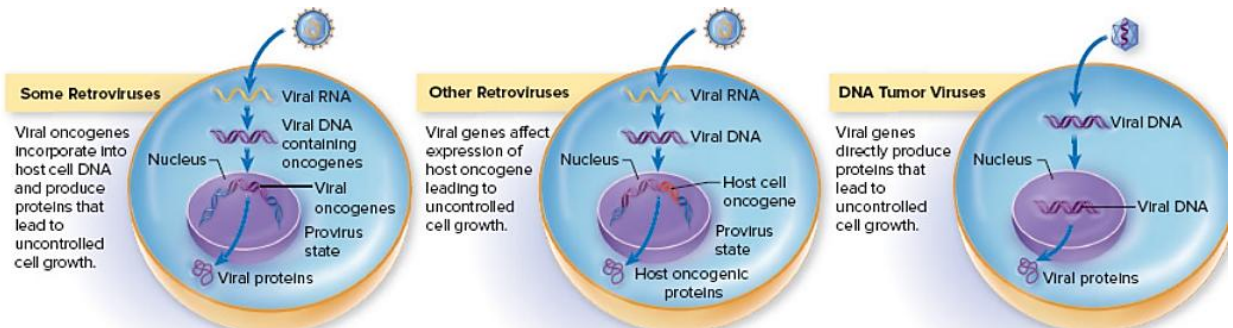
- Viruses that are strongly associated with human cancer.
- 8-13 % of human cancers are caused by oncogenic viruses.
- Seven viruses are classified as Group 1 carcinogen. Each group of viruses can cause various cancer
 - Human papillomavirus (mainly 16 & 18): increase risk for cervical and oropharyngeal cancer.
 - Epstein-Barr virus: increase risk for Burkitt's lymphoma.
 - HBV/HCV increased risk for HCC
 - Kaposi sarcoma-associated herpesvirus (KSHV), human T cell lymphotropic virus type-1 (HTLV-1), and human immunodeficiency virus (HIV).

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Multiple factors are for cancer viral oncogenesis in susceptible hosts:

- Insertion of viral genes in host genomes (provirus)
- Altered signaling pathways mediated by viral proteins.
- Chronic inflammation from persistent infection
- Inactivation of tumor suppressor gene (HPV can inactivate p53 genes).



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DNA virus

Adenovirus

Characteristics: Naked, icosahedral ds DNA virus

Presentation:

- Respiratory: Common cold characterized by cough, nasal congestion, fever, and sore throat. Lower respiratory tract infection may occur leading to pneumonia.
- Most common cause of viral conjunctivitis.
- GI leading to gastroenteritis and diarrhea.



Transmission: Aerosol droplets, Fecal-oral and contact with contaminated surfaces (virus is stable, can survive on surfaces).

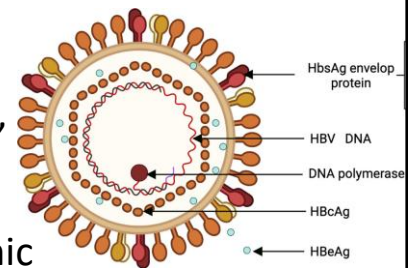
Treatment: Supportive treatment in most cases.

Hepatitis B virus

Characteristics: ds-gapped-DNA enveloped virus, family *Hepadnaviridae*.

Epidemiology: 300 million worldwide with chronic hepatitis B, >750 million annual mortality.

Transmission: Intimate physical contact with infected patient (semen). Encountering contaminated blood (bloodborne pathogen) or during birth. HBV can survive outside the body for few days.



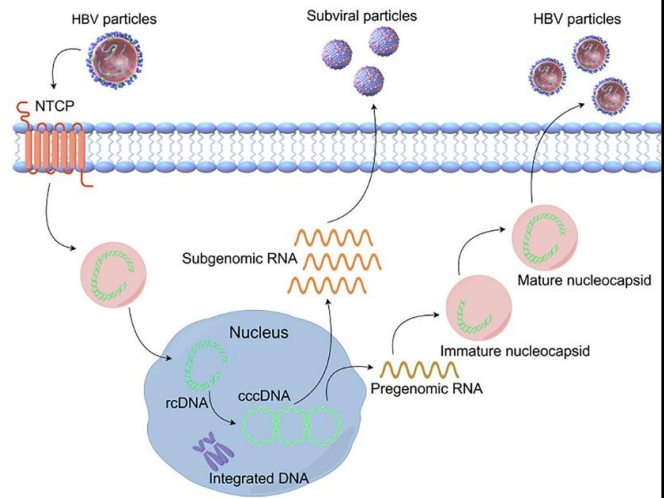
Clinical presentation: Hepatotropic virus, can cause acute and chronic liver disease (5-10 % of the cases).

- Acute hepatitis: fatigue, nausea, vomiting, jaundice, abdominal pain.
- Chronic hepatitis can progress to fibrosis, cirrhosis, liver failure and hepatocellular carcinoma (HCC).
- HBV has an incubation period around 90 days (between 1-6 month).

HBV life cycle

Unique features:

- 1- Viral DNA converts to cccDNA (covalently closed circular DNA). cccDNA is converted to mRNA with host transcriptase.
- 2- HBV is capable of chronic infection and has poor response to treatment due to stability of the cccDNA.
- 3- Can integrate in the host genome.
- 4- HBV DNA polymerase has reverse transcriptase activity that can synthesize a new DNA strand from the RNA template (unique, DNA virus replicate from mRNA template).



[Hepatitis B virus: modes of transmission, immune pathogenesis, and research progress on therapeutic vaccines](#)

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Diagnosis

A) Direct detection of viral DNA (viral load).

B) Antigens

- Hepatitis B surface antigen (HBsAg)
- Hepatitis B envelope antigen (HBeAg)

C) Antibodies

- Anti-hepatitis B surface antigen (Anti-HBsAg)
- Anti-hepatitis B envelope antigen (Anti-HBeAg)
- Anti-hepatitis B core antigen (Anti-HBcAg).

D) Elevated ALT/AST levels

Addressed after Immunology

- In the acute phase, you can detect the viral DNA, HBs antigen (appear within 1-2 weeks post infection) or HBe antigen (appear shortly after but clears quickly) and detect the IgM antibodies. In general, antibodies titer increase as acute infection resolves.
- In the chronic phase, anti HB core IgG and HBs antigen are present for >6 month. Anti HB core IgM is negative.
- Recovery “Window phase”: Anti HB core IgG antibodies
- Vaccinated: anti HBs IgG

Addressed after Immunology

DIAGNOSIS: ACUTE HEP B

→ ANTIGENS & ANTIBODIES

anti HB core IgM
ANTIBODIES

HBs



HBe



VIRAL DNA

HB core IgG
ANTIBODIES
↳ DEPENDS on TIME
of TESTING**DIAGNOSIS: CHRONIC HEP B**anti HB core IgM
ANTIBODIES

-

anti HB core IgG
ANTIBODIES

> 6 MOS.

HBs ANTIGEN



> 6 MOS.

VIRAL DNA



+/-

HBe ANTIGEN



+/-

HBe ANTIBODIES



+/-

RESOLVED HEPATITISanti HB core IgG
ANTIBODIES

+

anti HBs
ANTIBODIES

+/-

anti HBe
ANTIBODIES

+/-

ANTIGENS



-

anti HB core IgM
ANTIBODIES

-

VIRAL DNA



-

OSMOSIS.01

Addressed after Immunology

HEPATITIS B SEROLOGIC MARKERS

	VACCINATION	ACUTE HEPATITIS	RESOLVED ACUTE HEPATITIS	CHRONIC HEPATITIS
HBsAg	-	+	-	+
Anti-HBs	+	-	+	-
IgM anti-HBc	-	+	-	-
IgG anti-HBc	-	+	+	+
HB _e Ag	-	+	-	+/-
Anti-HB _e	-	+	+	+

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Treatment and prevention:

(1) Prevention: vaccination against hepatitis B.

- Infants and children: 3 doses given over a 6-month period (2nd dose, 1 month after 1st dose and 3rd dose, 6 months after 1st dose).
- Adults (>18 y): 2 doses, 1 month apart.

(2) Treatment are either supportive or interferon and other antivirals to decrease the viral replication. No definitive cure for HBV.

RNA viruses

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Hepatitis A and Hep. E viruses

Characteristics: ssRNA virus (+ve sense), non-enveloped

Transmission: Fecal-oral route

Cause: Acute self-limiting hepatitis

Symptoms: Fever, fatigue, nausea, vomiting, jaundice, abdominal pain

Treatment: self limiting. Supportive management

Hep A prevention: Hepatitis A vaccine

HEV

- HEV generally causes self-limiting viral hepatitis.
- In pregnant females, HEV infection can be severe.
 - Associated with up to 30% mortality in the third trimester due to fulminant hepatitis.
 - Associated with high rates of preterm labor and increase intrauterine death.
 - Increase vertical transmission of HEV.

Hep D virus

(-)ssRNA enveloped

Only infect individuals with active HBV. HDV lacks genes to make its envelope. Needs HBV surface antigen to make its envelope.

Transmission: sexual and blood

Diagnosis: Hep. D RNA, delta antigen or anti-hepatitis D antibodies

Presentation:

- Co-infection of HBV and HDV can increase severity of liver symptoms.
- Superinfection: Hep D on a chronic Hep B infection. Increase risk of fulminant hepatitis (up to 40% of fulminant hepatitis)

Treatment/prevention: manage/prevent HBV will prevent HDV

Hepatitis C virus

Characters: enveloped (+) ss RNA, flavivirus family. High degree of genetic variation. Seven genotypes (USA: 1a, 1b, 2 and 3).

Transmission: contaminated blood (bloodborne pathogen) or sexual contact (uncommon for HCV).

Diagnosis: Serological exams to determine viral antibodies (7-31 weeks of infection) or antigens (1-3 weeks post exposure) and PCR to detect the viral RNA for confirmation.

Clinical presentation: Haptotropic, causing acute or **chronic liver infection**. > 50 % cases develop chronic infection. Chronic infection can lead to cirrhosis, liver failure/hepatocellular carcinoma.

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Treatment:

HCV direct-acting antiviral agents (DAAs) against the viral structural and non-structural proteins. Consequently, this affects several steps in the virus life cycle (like, NS3/NS4A protease inhibitors; Harvoni®) with SVR>90 %.

Peginterferon and ribavirin combination, SVR around 40%.

Ribavirin is effective against a broad spectrum of RNA and DNA viruses. Also, it can be used in combination with other DAAs.

Viral hepatitis

Virus	Genome/envelope	Family	Transmission	Vaccine
HAV	ssRNA/No Env.	Picornavirus	Fecal-oral	Yes
HBV	dsDNA/ Env.	Hepandavirus	Sexual and blood	Yes
HCV	ssRNA/ Env	Flavivirus	bloodborne	No
HDV	ssRNA/Env	Deltavirus	Sexual and bloodborne	No
HEV	ssRNA/no Env	Calicivirus	Fecal-oral	No

[The ABCs of Hepatitis – for Health Professionals \(cdc.gov\)](#)

[Hepatitis | Osmosis](#)

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Respiratory syncytial virus

Characteristics: ss RNA (-ve sense), enveloped virus. Contain its own polymerase RNA-dependent RNA polymerase. Replicates in nasopharynx.

Transmission: Respiratory droplets and direct contact.

Presentation: Common respiratory infection in Winter. Fever, coughing, rhinitis, pharyngitis. Symptoms are more severe in infants and immunocompromised. We can get recurring throughout life due to viral mutations (antigenic drift)

Retroviruses

1) The oncovirus group, which contains the sarcoma and leukemia viruses (e.g., human T-cell leukemia virus [HTLV]).

2) The lentivirus (“slow virus”) group, includes human immunodeficiency virus (HIV).

- Unique: reverse transcribes its RNA genome into DNA by a viral reverse transcriptase enzyme.
- The produced viral DNA goes to the nucleus and integrates a DNA copy of the RNA genome into cellular DNA. Only retroviruses perform this step.
- Integration is an obligatory step in viral replication.

HIV Infection and AIDS

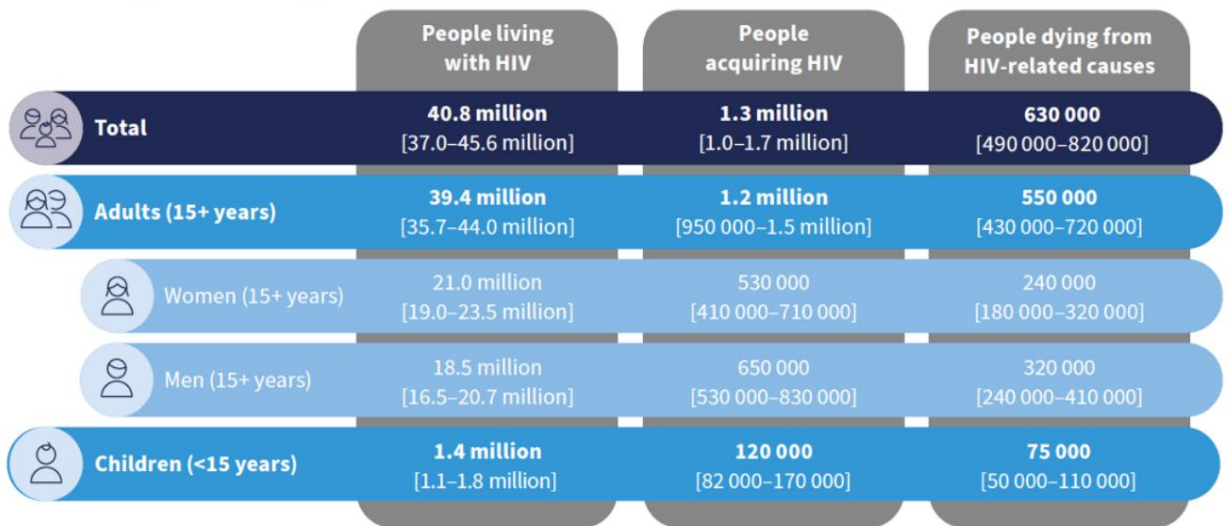
Virus: Human Immunodeficiency Virus (HIV)

Transmission: Sexual contact, blood/blood products and vertical (maternal-child).

Presentation: Acute and chronic HIV infection. If left untreated, can lead to acquired immunodeficiency syndrome(AIDS) which will be fatal due to opportunistic infections.

Management/prevention: No cure, no vaccine but can be treated and controlled by life-long antiviral medications.

Summary of the global HIV epidemic, 2024



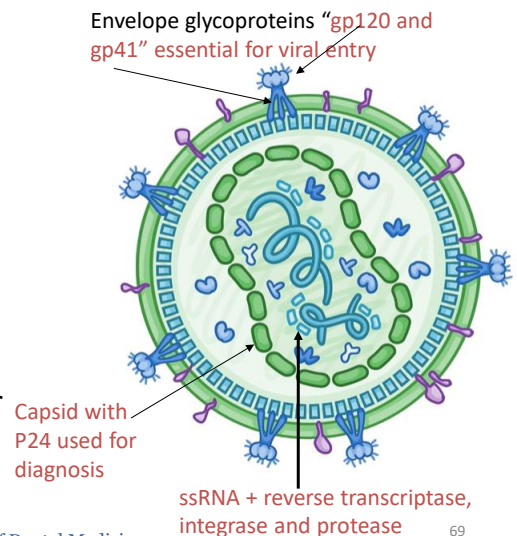
[U.S. Statistics | HIV.gov](#)
[Global Statistics | HIV.gov](#)

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Human Immunodeficiency virus

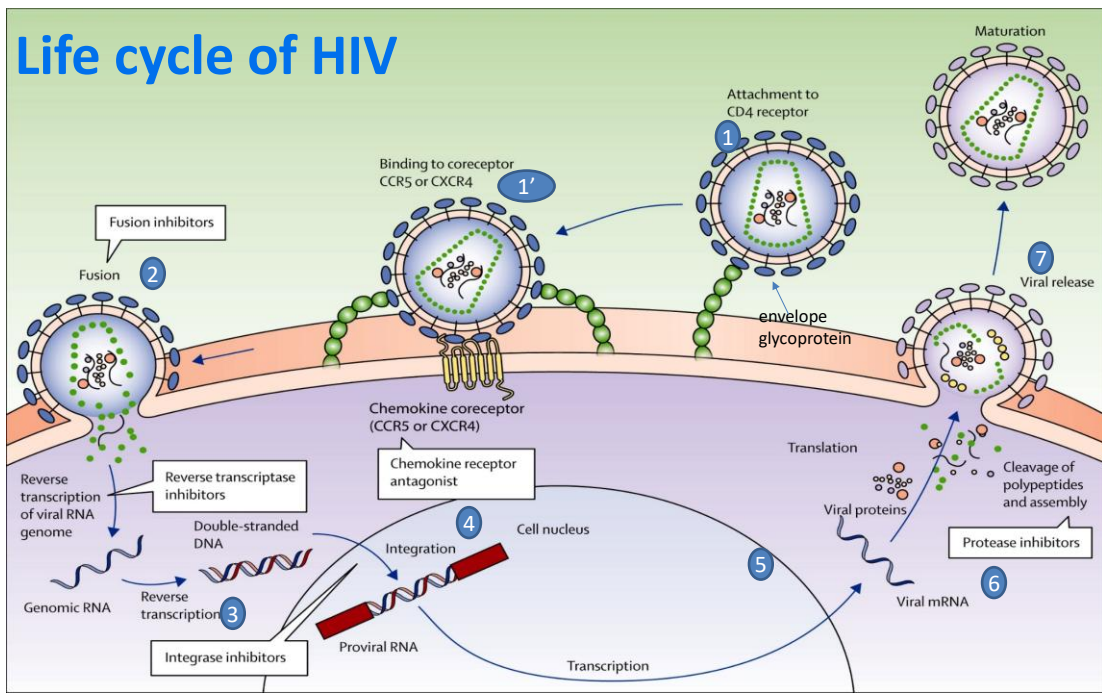
- Enveloped ss (+) sense RNA
- Long incubation period preceding symptom development.
- Immune system fails to mount sterilizing immune response. Infection by HIV may lead to acquired immunodeficiency syndrome (AIDS).
- HIV preferentially infects helper (CD4) T lymphocytes. In addition, HIV can infect other cells (macrophages and dendritic cells) that have CD4 proteins on their surfaces.



Subtypes

	HIV-1	HIV-2
Virulence and transmission	High. Majority of patients progress to AIDS	Lower virulence and lower transmission. Most infections do not progress to AIDS.
Diversity	High (subdivided to 4 groups M, N, O and P; group M is subdivided into 9 subtypes)	Low
Prevalence	Global	West Africa
Vertical transmission (mother to fetus)	Present	Rare

Life cycle of HIV



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1- Binding: The interaction between the HIV envelope glycoprotein (gp120) with the host cell CD4 surface receptor marks the initiation of the infection cycle.

2- Fusion or entry: Binding of the viral glycoprotein to the chemokine receptors CCR5 (R5-tropic) or CXCR4 (X4-tropic), co-receptors, initiates the membrane fusion and viral entry. CCR5 is present on macrophage lineage (M tropic) and T cells. While, CXCR4 is found mainly on T cells (T-tropic). HIV can also have dual tropism.

3- Reverse transcription: HIV is a single stranded RNA virus. Inside the cell, the viral RNA is converted to DNA by reverse transcriptase enzyme. Unique for retroviruses.

4- Integration: retroviral integrase enzyme allows the integration of the viral DNA into the host DNA

5- Replication: The viral DNA is replicated using the host cell machinery to yield viral RNA and proteins.

6- Assembly: The viral genetic material and capsid protein are assembled near the membrane. Aspartate protease leads to the cleavage of viral polyproteins and finally assembly of an infectious viral particle.

7- budding: This stage refers to the release of new viral molecules outside the cells.

Disease Course

Acute infection “Stage 1 or acute retroviral syndrome”

- Within days after infection. Last for few weeks. Increased virus titer. Flu-like symptoms (fever, fatigue, soar throat night sweats) or GI symptoms.

Chronic HIV infection (Stage 2)

- Mostly asymptomatic. Patients can transfer the disease.
- Latent stage that can last for a decade.
- During this stage, there is a steady decline in the T-cell count along with steady increase in the viral load. CD4 T helper are imp. for adaptive immune response and activation of neutrophils, macrophages and cytotoxic T-cells.
- Patients can develop “AIDS-related complex (ARC) syndrome”=symptomatic infection without AIDS stage.
- Most frequent manifestations are persistent fevers, fatigue, weight loss, and **persistent lymphadenopathy**.

AIDS Stage (stage 3)

Most severe stage of HIV:

- Natural course of untreated disease.
- CD4 count < 200/ul. Infected T cells are killed by the cytotoxic cells and natural killer cells.
- High viral load.
- HIV infection dysregulates the T-cell differentiation and causes the autoimmune destruction of infected T cells.
- Generalized lymphadenopathy, fever, weight loss, and chronic diarrhea
- Patients suffer from multiple opportunistic infections (fungal, bacterial, viral infections) and cancer.

Staging of HIV

CDC HIV Infection Stages 1–3 Based on Age-Specific CD4+ T Lymphocyte Count or CD4+ T Lymphocyte Percentage of Total Lymphocytes^a

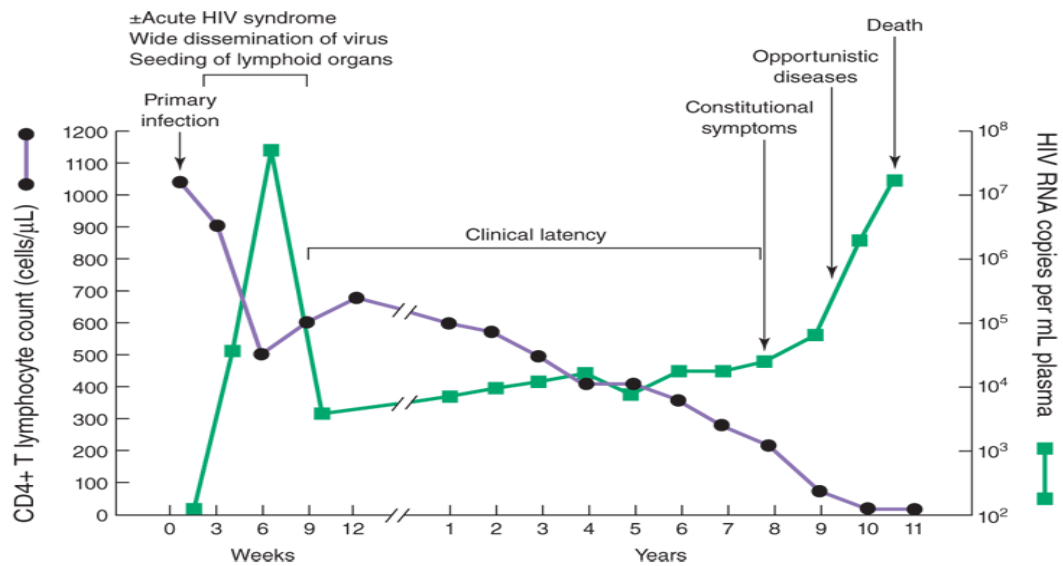
	AGE ON DATE OF CD4 T+ LYMPHOCYTE TEST					
	<1 YEAR		1–5 YEARS		6 YEARS THROUGH ADULT	
STAGE ^a	CELLS/μL	%	CELLS/μL	%	CELLS/μL	%
1	≥1500	≥34	≥1000	≥30	≥500	≥26
2	750–1499	26–33	500–999	22–29	200–499	14–25
3	<750	<26	<500	<22	<200	<14

^a The stage is based primarily on the CD4+ T lymphocyte count; the CD4+ T lymphocyte count takes precedence over the CD4+ T lymphocyte percentage, and the percentage is considered only if the count is missing.

For Illustration

California Northstate University College of Dental Medicine

Typical course of HIV infection



Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition
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Lab parameters



- 1- Serology: Test for P24 antigen and anti-HIV antibodies.
 - 2- HIV-1/HIV-2 antibody differentiation assay.
 - 3- Nucleic acid test (NAT). This test detects viral genetic material via RT-PCR.
- **CD4 count for T cells**
 - Normal range: 500 -1400 cells/ul. CD4 count >500ul equivalent to a CD4 percentage of >29 %.
 - CD4 count between 200 to 500 cells/ul, corresponds to a CD4 percentage of 14 to 28 %.
 - CD4 count <200 cells/ul corresponds to a CD4 percentage of <14 %. AIDS defining condition.
 - **ANC “absolute neutrophil count”** (normal 1500-8000/ul). ANC <1000/mm³ indicates a significant increase of risk for infection.

Treatment

Patients with HIV infection are administered the highly active anti-retroviral therapy (HAART). This is a customized combination of different drug classes that target different steps of the viral life cycle.

The combination will be based on:

- viral load,
- HIV strain,
- CD4 cell count
- Patient condition and other disease symptoms.

Classes of antiretroviral agents:

1- Entry inhibitors

- a) Fusion inhibitors (FIs)
- b) Chemokine receptor antagonists (CCR5 antagonists).

2- Reverse transcriptase inhibitors:

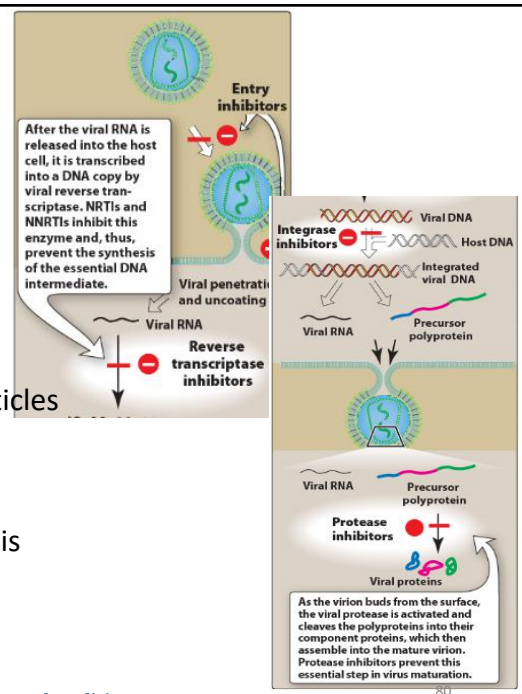
- a) Nucleoside reverse transcriptase inhibitors (NRTIs)
- b) Non-nucleoside reverse transcriptase inhibitors (NNRTIs).

3- Integrase inhibitors (INSTIs).

- 4- **Protease inhibitors (PIs).** Reversible inhibitors of HIV aspartyl protease. Prevent the maturation of viral particles leading to the production of noninfectious virions.

HIV-2 treatment is similar to HIV-1 treatment.

One exception “NNRTIs are not used because HIV-2 virus is resistant to this class “.



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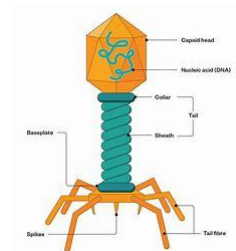
Prevention

- No vaccine available.
- Patient education and safe practice
- Post exposure prophylaxis
 - Significantly reduces the risk of getting HIV.
 - taking anti-retroviral regimen within 72 hours after a possible exposure to prevent HIV. Mediations are taken everyday for 28 days with lab diagnosis at 6 w, 3m and 6m.

The good that viruses do !

- In 2015, the first **oncolytic virus** therapy based on herpesvirus was approved by the FDA and European Medicines Agency for the treatment of melanoma lesions in the skin and lymph node. Oncolytic viruses can replicate selectively in cancer cells and eventually lyse and kill those cells without harming the non-tumor cells. Therefore, selective killing can offer a novel approach to cure cancer.
- Attenuated viral vectors with specific viral antigens can be one of the vaccine strategies. Those include attenuated adenovirus, measles virus or vascular stomatitis virus (VSV).

- **Bacteriophage** is a virus that can infect bacteria in a species-specific manner. Infects bacterial in a lytic or lysogenic (integration) replication manner. Possible future use: provide antimicrobial defense that actively protects mucosal surfaces.



- **Adenovirus vectors and lentiviruses** Used in targeted gene therapy. To restore the function of a defective gene, those vectors ensure the delivery and expression of therapeutic genes in the desired target cells. Adenovirus vector for the treatment of lipoprotein lipase deficiency was approved by the European Medicines Agency in 2012.

Resources

